Analysis of Factors Driving Carbon Market Prices

Undergraduate Thesis – School of Finance, SUFE

Xiangzhong Meng*

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Abstract

This paper studies the development and pricing dynamics of carbon markets. In the global market, carbon prices are pro-cyclical with carbon allowance taking domination. In China, market premium and industrial factors (metallurgical industry, aviation, and broad agriculture) do explain overall allowance returns, but the regional effect is significant. In regional exchanges, price determinants show stronger negative correlations with energy prices. In national exchanges, returns are more influenced by industry specifics than by market premium.

Keywords: Carbon market, Carbon Allowance, Regional Effect, Asset Pricing

*Imperial College London, Email: xiangzhong.meng24@imperial.ac.uk

1 Introduction

1.1 Research Background

With the rapid growth in per capita GDP, global carbon dioxide emissions increased by 57.7% from 21.28 billion tons in 1990 to 33.57 billion tons in 2020.[1] Under the pressure of environmental, ecological, and economic challenges, the UNFCCC was adopted in 1992 as the first international treaty addressing global climate change. The Kyoto Protocol (1997) introduced carbon trading mechanisms, allowing countries to trade emission allowances or invest in emission-reduction projects in developing countries.

Carbon markets offer economies the ability to price carbon emissions directly. Carbon taxes, as a supplementary policy tool, help increase fiscal revenue, ensure policy enforcement, promote public acceptance, and address global competitiveness. Between 2013 and 2023, the coverage and revenue of global carbon pricing instruments steadily increased. As of 2022, 73 carbon pricing mechanisms covered 23% of global GHG emissions, with total transactions exceeding $100 billion.

China, as a major emitter and signatory to the Kyoto and Paris Agreements, initiated pilot ETS programs in 7 provinces in 2011. Shenzhen's carbon market launched in 2013, followed by the national ETS in 2021, which now covers 8.92% of global GHG emissions - approximately 23% of all combined countries.[2]

To meet climate commitments, it is vital to maintain or raise carbon prices. However, economic, political, and environmental uncertainties pose challenges. Between 2023 and 2024, global factors like unconventional monetary easing, geopolitical tensions, and energy transitions disrupted carbon pricing. These shifts highlight the importance of scientific carbon pricing models to guide

[1] Data source: World Bank

[2] Data source: NDRC document, Notice on Launching Pilot Carbon Emission Trading Schemes (NDRC Climate [2011] No. 2601)

sustainable development.

1.2 Why researching is important

Compared to traditional financial instruments, carbon products have complex pricing mechanisms. They vary in type (taxes, ETS, voluntary offsets), usage, and institutional structure. This paper systematizes the characteristics of carbon tools globally and in China to illuminate market structures and dynamics.

Carbon pricing is influenced by energy supply-demand, economic cycles, industry performance, and local factors. We explore these drivers using empirical data from international markets and China's 9 carbon exchanges, offering a foundation for future research.

For China, as the largest developing country, scientific carbon pricing is vital to align policy guidance with sustainability. Our empirical study provides quantitative insights into price drivers across different sectors, offering implications for both policy and academia.

1.3 Research Method and Structure

This study applies empirical methods, utilizing international carbon price data (from World Bank) and Chinese ETS price and financial market data (from CSMAR). We perform time-series and regression analyses to uncover macroeconomic, industrial, and regional factors influencing carbon prices.

The paper is structured as follows: Section 2 reviews the literature; Section 3 introduces the data and empirical setup; Section 4 presents regression results and robustness checks; Section 5 concludes with key findings and policy recommendations.

1.4 Contributions

This study makes the following contributions:

- It systematically summarizes the pricing and efficiency characteristics of different carbon tools, enabling clearer understanding of the fragmented global carbon market.
- It empirically investigates drivers of both international and domestic carbon prices, incorporating environmental, industrial, and macroeconomic dimensions.
- It disaggregates Chinese ETS price drivers by regional exchange and covered industries, refining understanding of micro-level impacts.

2 Literature Review

2.1 Heterogeneity in Carbon Pricing Mechanisms

Carbon pricing varies in type, regulation, and effectiveness. According to the World Bank (2023), pricing tools include ETS, carbon taxes, crediting mechanisms, results-based climate financing (RBCF), and internal pricing. Markets are classified as compliance, voluntary, or offset.

ETS prices derive from marginal abatement cost theories, though scholars such as Montgomery (1972) and Luo (2024) highlight how transaction costs and surplus allocations cause price deviations. Regional systems show heterogeneity due to overlapping jurisdictions and fragmented governance structures (Jiang and Yu, 2024).

2.2 Implications of Mechanism Design

Zhen et al. (2023) and Lin and Li (2010) argue that developed nations' hybrid ETS-tax schemes intensify green trade barriers, curbing developing countries' exports. Pan and Yang (2024) emphasizes regressive impacts and welfare losses from carbon taxes. Wang and Yang (2021) finds China's regional pilot ETS lowers carbon-finance market linkages, while Liu and Xu (2023) reports positive spillovers to neighboring provinces using DID models.

2.3 Domestic Drivers of Carbon Prices

Zhang et al. (2021) argue that China's ETS, unlike cap-based systems abroad, is output-intensity-based and combines taxes/subsidies. He (2011) links permit allocations to governmental emission goals. Xu and Lien (2024) documents strong spillovers between ETS returns and inflation. Li (2023) finds exchange rates and industrial production influence prices, while Wang and Xu (2021) explores volatility from tech shocks and policy interventions.

2.4 International Drivers of Carbon Prices

Seifert and Uhrig-Homburg (2008) eliminate seasonality in price models. Burger et al. (2008) identify oil prices, political risk, temperature, development, and tech advancement as key drivers. Mesrzade et al. (2023) show supply chain emissions impact firm-level pricing. Liu and Xu (2023) use interval decomposition to quantify COVID and Ukraine war shocks on EU ETS.

3 Empirical Preparation

3.1 Carbon Markets and Products

3.1.1 Compliance Markets and Emission Trading

Compliance markets consist of primary and secondary markets. In the primary market, government bodies allocate emission permits with caps. In the secondary market, permit holders trade based on their needs.

Globally, 29 compliance markets exist (e.g., EU ETS, UK ETS, Korea, California). China's ETS includes both national and provincial systems. The national market launched in 2021 focuses on the power sector, while local pilots (e.g., Shenzhen, Beijing, Guangdong) span broader industries.

3.1.2 Voluntary Markets and Offset Credits

Voluntary carbon markets (VCMs) allow businesses or individuals to buy carbon credits from projects reducing or removing emissions. These credits are verified by third parties like Verra or Gold Standard.[3]

Credits may stem from:

- **Emission reductions or avoidance:** e.g., renewable energy, improved efficiency
- **Removals and sequestration:** e.g., afforestation, direct air capture

3.1.3 Carbon Taxes and Border Adjustments

Carbon taxes set a fixed price on emissions and typically supplement ETSs. Although not yet implemented, China's 2022 green consumption plan proposes carbon taxation.

Carbon border adjustment mechanisms (CBAMs), like the EU's 2026 plan, aim to avoid carbon leakage by imposing tariffs on imported emissions-intensive goods (e.g., steel, cement, aluminum).

3.2 International Data on Carbon Pricing

3.2.1 World Bank Data Sources

This study uses World Bank dashboards [4] for carbon tax and ETS coverage, prices, and GDP data from 2005 to 2022. These datasets are widely cited and follow statistical best practices.

[3] VERRA was established in 2005 through a joint initiative by the Climate Group (CG), the International Emissions Trading Association (IETA), and the World Economic Forum (WEF). It operates the Verified Carbon Standard (VCS) program and is currently one of the most influential organizations in the voluntary carbon market.

[4] See official World Bank source: go.worldbank.org

3.2.2 Coverage and Pricing Trends

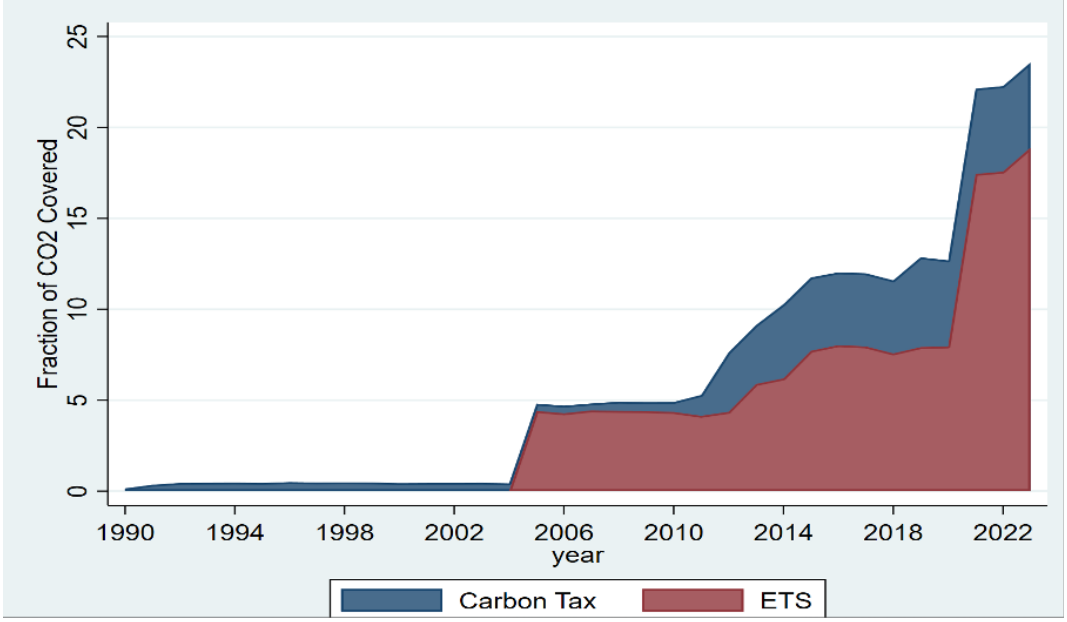


Figure 1: Global Carbon Coverage by Mechanism Type (2005–2022)

As is shown in Figure 1, ETSs cover more GHGs than carbon taxes globally. Coverage rose from 4.41% (2005) to 18.88% (2022). Rapid gains post-2019 stem from system launches in China and Vietnam.[5]

[5]The national carbon exchange in China was launched in 2021, followed by the establishment of Vietnam's national exchange in 2022. Trading volume data is available from the World Bank's Carbon Pricing Dashboard: carbonpricingdashboard.worldbank.org.

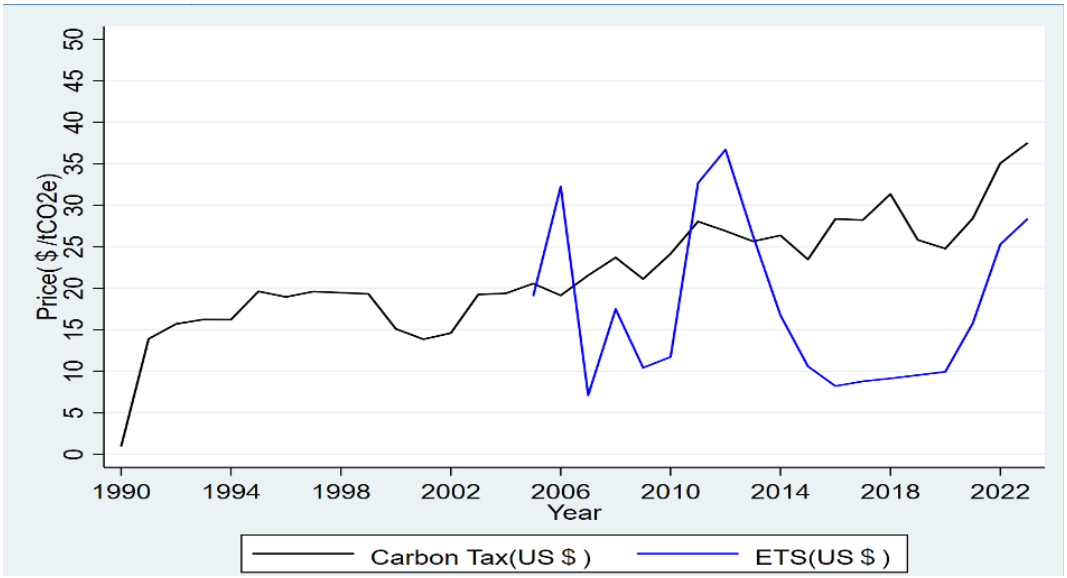


Figure 2: Global Carbon Pricing by Mechanism Type (2005–2022)

As is shown in Figure 2, price levels are calculated by weighting country-level carbon prices by their GHG coverage.[6] ETS prices exhibit high volatility and fragmentation. Carbon tax prices are steadier.

[6]The weighted average carbon price is calculated as:

$$p_i = \sum_k w_k \cdot p_k$$

where w_k denotes the percentage share of global greenhouse gas emissions covered by mechanism k in year i, and p_k is the corresponding carbon price.

3.2.3 Cyclicality of Carbon Prices

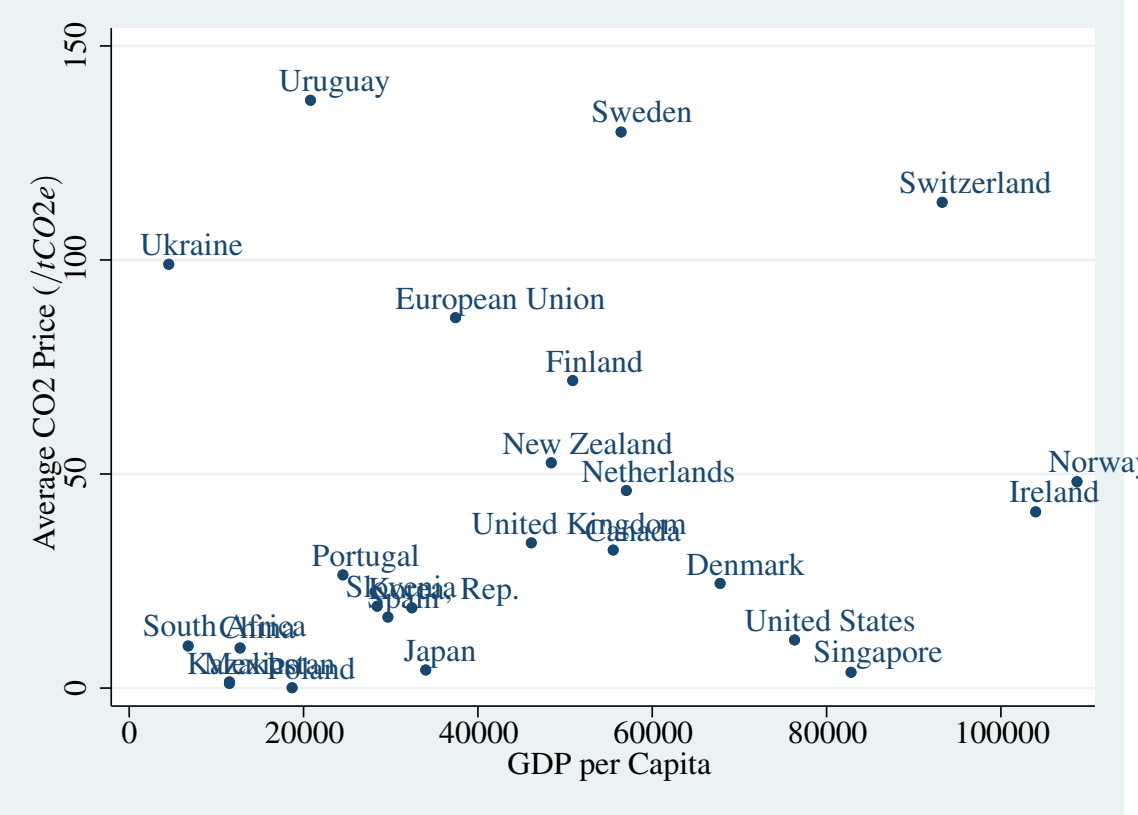


Figure 3: Carbon Market Prices across Nations

Plotting 2022 per-capita GDP vs. carbon prices reveals a positive correlation as Figure 3, suggesting pro-cyclicality. Outliers like Ukraine (high price, low GDP) reflect geopolitical disruptions. Developed countries like Norway show low carbon prices despite high GDP due to clean energy penetration.

3.3 China's ETS Data and Setup

3.3.1 Data Sources

Chinese ETS and market indices are obtained from the CSMAR database. The study uses daily price and volume data from 2013–2024 across 9 ETS platforms.

3.3.2 Market Coverage

Industry	BJ	TJ	SH	CQ	HB	GD	SZ	FJ	Nat
Electric Power	✓						✓		✓
Steel		✓	✓	✓	✓	✓	✓	✓	
Building Materials	✓	✓	✓	✓	✓	✓	✓		
Petrochemical	✓	✓	✓	✓	✓	✓		✓	
Chemical Industry		✓	✓	✓	✓		✓	✓	
Non-ferrous Metals		✓	✓	✓	✓		✓	✓	
Paper Making				✓	✓	✓	✓	✓	
Aviation	✓	✓	✓			✓			✓
Transportation	✓		✓				✓		
Construction			✓						
Other Industries	✓	✓	✓	✓	✓		✓	✓	
Waste Treatment				✓				✓	
Food and Beverage		✓		✓	✓				
Service Industry	✓		✓				✓		

Table 1: Industry Coverage Across Regional ETS Platforms

As Table 1 shows, local ETSs (e.g., Shenzhen, Guangdong) cover multiple sectors including power, steel, chemicals, aviation, and cement. The national ETS covers only the power sector but dominates in volume since 2021.

3.3.3 Time Series Behavior

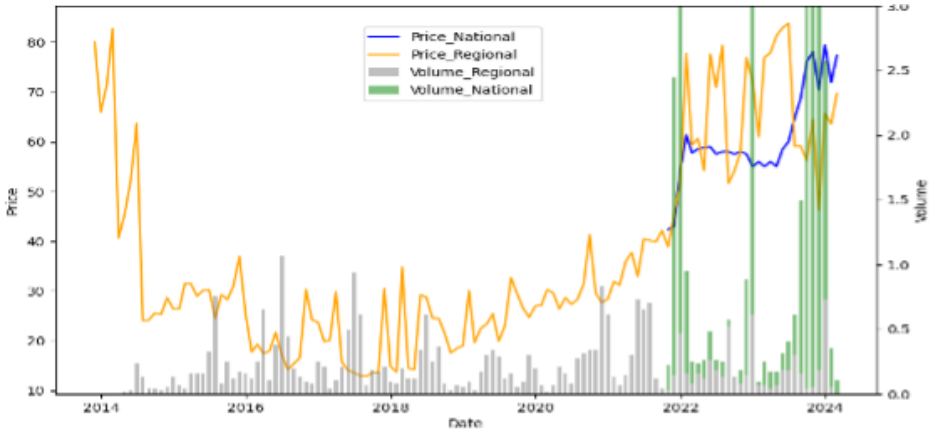


Figure 4: China Monthly Carbon ETS Trading Price and Volume

Year	Mean	Standard Deviation
2016	0.26	6.72
2017	0.11	5.24
2018	0.13	6.25
2019	-0.05	5.64
2020	0.09	5.26
2021	0.20	6.00
2022	0.20	3.06
2023	0.02	2.18
2024	-0.12	2.47

Table 2: Summary Statistics of China's Carbon Emission Allowances (2016–2024)

As shown in Figure 4, carbon prices in China's regional pilot exchanges—led by Shenzhen—were abnormally high before 2014, likely due to low market liquidity. Over time, as trading volumes increased across pilot platforms, carbon prices gradually rose. Notably, following the establishment of the national carbon emission trading scheme in 2021, the national exchange replaced local

pilots as the dominant trading venue. However, it did not crowd out provincial exchanges; rather, it stimulated both price levels and transaction volumes in the pilots. This trend was particularly pronounced between 2022 and 2023.

Regarding price volatility, the overall volatility of carbon allowances declined from 2018 to 2021, in line with increasing market depth (see Table 2). From 2022 to 2024, volatility exhibited a distinct seasonal pattern. Specifically, trading volumes typically fell during mid-year months, while rising in late-year and early-year periods (November to February), accompanied by more pronounced price fluctuations. In late 2023, an unexpected spike in prices at the national exchange temporarily drove down the aggregate carbon price across platforms, a divergence that gradually faded in early 2024.

In summary, both transaction volume and pricing levels in China's carbon market show an upward trend. The national and regional markets display distinct characteristics. For empirical purposes, the 2018–2024 period provides a stable window in which carbon prices and volatilities appear smoother and more suitable for econometric analysis.

3.3.4 Candidate Price Drivers

This study proposes several categories of explanatory variables that may influence carbon allowance prices, each supported by theoretical justification.

First, macroeconomic variables are considered, both domestic and international. Domestic indicators are divided into stock and bond indices: stock indices reflect the general business cycle, corporate earnings, and investment activity, while bond indices provide information on prevailing credit spreads and market liquidity. This distinction allows for a more nuanced understanding of the financial determinants of carbon asset pricing. International macroeconomic conditions may affect global commodity prices, foreign exchange rates, and other relevant financial variables.

Second, energy price variables are included. Carbon trading schemes primarily cover fossil fuel-

based sectors, including coal, oil, natural gas, and LPG. However, the futures markets for these energy types in China vary in terms of inception dates and data availability. To ensure temporal consistency and avoid missing data, we employ a broad-based energy futures index and a coal futures index—both of which exhibit continuous coverage and stable volatility over the sample period.

Third, we consider industry-specific performance as a potential driver. Based on Table 1, which outlines the industrial coverage across China's carbon trading platforms, and Figure**??**, which summarizes price and volume trends, we identify three representative sector indices: agriculture, metallurgy, and aviation. According to the International Energy Agency and Statista data, the industrial sector accounted for approximately 30% of China's total emissions in 2020 (excluding energy production), while aviation and agriculture contributed 8% and 1%, respectively. Given their significance in carbon emissions and their relevance to ETS coverage, these sectors are included as explanatory variables.

Lastly, we incorporate regional market segmentation. Given structural differences between the national and local ETS platforms—including variation in industry coverage, trading volume, and regulatory design—we conduct separate regressions for the national and provincial ETS prices. This approach enables us to examine whether macroeconomic, energy, and sectoral factors exert different effects across distinct carbon market regimes.

4 Empirical Results

4.1 Model Construction

This section describes the data sources, preprocessing steps, and the empirical model used to estimate the determinants of carbon allowance excess returns in China.

4.1.1 Data Selection and Processing

We collect daily data from the CSMAR database covering the period from 2018 to 2024. The variables used in this study include:

- China's carbon allowance prices,
- China Securities 500 Index (CSI500),
- S&P 500 Index (SPX),
- ChinaBond Composite Bond Index (Debt),
- CSI Metals Futures Index (Metal),
- CSI Agriculture Index (Farm),
- CSI Airline Industry Index (Airline),
- CSI Energy Futures Index (Energy),
- CSI Coal Futures Index (Coal),
- Daily risk-free interest rate.

We calculate daily excess returns for each index by subtracting the risk-free rate from the raw return. These excess returns are aligned to trading days and used for regression analysis.

4.1.2 Model Specification

Based on the variables above, the following linear regression model is constructed:

$$
\begin{aligned}
Carbon_t = \beta_1 \cdot CSI500_t + \beta_2 \cdot SPX_t + \beta_3 \cdot Debt_t \\
+ \beta_4 \cdot Metal_t + \beta_5 \cdot Airline_t + \beta_6 \cdot Farm_t \\
+ \beta_7 \cdot Energy_t + \beta_8 \cdot Coal_t + \varepsilon_t
\end{aligned} \tag{1}
$$

where:

- $Carbon_t$: Excess return of China's carbon allowances on day t,
- $CSI500_t$: Excess return of the CSI500 Index,
- SPX_t: Excess return of the S&P 500 Index,
- $Debt_t$: Excess return of the composite bond index,
- $Metal_t$: Excess return of the metals futures index,
- $Airline_t$: Excess return of the airline industry index,
- $Farm_t$: Excess return of the agriculture index,
- $Energy_t$: Excess return of the energy futures index,
- $Coal_t$: Excess return of the coal futures index,
- ε_t: Noise term.

This model allows us to investigate the influence of macroeconomic, industrial, and energy market factors on carbon allowance pricing.

4.1.3 Descriptive Statistics

Table 3 summarizes the main statistics:

Variable	Mean	Std. Dev.	Min	Max
Carbon	0.15	3.80	-17.03	19.78
CSI500	-0.05	1.09	-5.49	8.98
SPX	0.02	0.96	-2.98	2.47
Debt	0.02	0.04	-0.08	0.09
Metal	0.00	0.78	-1.61	1.54
Airline	0.04	1.41	-2.83	2.77
Farm	-0.06	1.26	-4.92	4.19
Energy	0.02	1.30	-2.43	2.55
Coal	0.01	1.47	-6.35	5.65

Table 3: Descriptive Statistics of Key Variables

The descriptive statistics presented in table 3 provide valuable insights into the behavior of the variables used in this study. The excess return on carbon allowances (*Carbon*) exhibits considerable variability, with a mean of 0.15 and a standard deviation of 3.80. This high level of dispersion suggests the presence of extreme values and reflects heterogeneity across different trading venues and industry sectors.

The CSI500 Index demonstrates relatively stable characteristics, with a mean close to zero (-0.05) and a modest standard deviation of 1.09. This indicates relatively low market volatility and a steady trend, potentially shaped by macroeconomic fundamentals.

Other variables, including the Coal Futures Index (*Coal*), the Agriculture Index (*Farm*), the S&P 500 Index (*SPX*), the Airline Index (*Airline*), and the Energy Futures Index (*Energy*), display unique statistical features. For instance, the coal and agricultural indices have near-zero mean

returns (0.01 and -0.06, respectively) but large standard deviations, implying significant intra-sector variation. The SPX also shows a mean close to zero and a relatively low standard deviation, suggesting a stable global market environment.

In contrast, the airline and energy sectors show greater volatility, with standard deviations of 1.41 and 1.30, respectively. This may be driven by macroeconomic shocks and sector-specific factors such as fuel prices or demand cycles.

Finally, the *Debt* (ChinaBond Composite Index) and *Metal* (CSI Metals Futures Index) variables appear comparatively stable. The mean and standard deviation for *Debt* are 0.02 and 0.04, respectively, while those for *Metal* are 0.00 and 0.78. These values suggest relatively mature and liquid markets with lower levels of short-term price fluctuation.

4.2 Regression Results

4.2.1 Full-Sample Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
CSI500	0.127**								0.061
	(2.29)								(0.75)
Debt		3.817***							4.439***
		(2.82)							(3.24)
Metal			0.141*						0.084
			(1.83)						(0.85)
Energy				0.039					-0.142
				(0.85)					(-1.57)
Airline					0.073*				0.056
					(1.71)				(1.08)
Farm						0.136***			0.112*
						(2.83)			(1.83)
Coal							0.062		0.083
							(1.52)		(1.05)
SPX								0.060	0.060
								(0.94)	(0.94)
N	3954	3954	3954	3954	3954	3954	3954	3954	3954
Adj. R^2	0.001	0.002	0.001	-0.000	0.000	0.002	0.000	-0.000	0.004

Note: This table reports stepwise OLS regression estimates. T-statistics are reported in parentheses. $*$ $p < 0.1$, $**$ $p < 0.05$, $***$ $p < 0.01$.

Table 4: Full-Sample Regression Results

This section presents linear regressions for each of the eight explanatory variables against the excess return of carbon allowances. The analysis is conducted using the full sample, which includes both national and regional (non-national) exchange data. The results are summarized in Table 4.

Among the full sample regressions, five explanatory variables exhibit statistically significant coefficients when regressed individually: the CSI500 Index, the ChinaBond Composite Index, the Metals Futures Index, the Airline Industry Index, and the Agriculture Index. Notably, the bond and agriculture indices remain significantly and positively associated with carbon excess returns in the full model, at the 1% and 10% confidence levels, respectively. The significance of the remaining variables diminishes when all controls are included.

Compared to the relatively weak explanatory power of the domestic equity market in the full model (despite 5% significance in the individual regression), bond market conditions exert a more robust positive effect on carbon pricing. This may be due to the fact that an increase in bond yields signals higher credit costs for firms, thereby increasing the financing cost of carbon transactions and elevating returns on carbon allowances. In contrast, equity returns reflect broader market sentiment and investment activity, which may be less directly linked to transaction-level costs in the carbon market.

Among sectoral variables, agriculture is the only industry that retains significance in the full regression (10% level). One plausible explanation is that agricultural activities encompass a broader range of emission sources compared to other industries such as aviation or metallurgy. These include biological respiration, fertilizer and pesticide use, and carbon emissions from food processing. While the airline industry shows marginal significance in the individual regression (10% level), this disappears in the full model. This suggests that the original signal may be driven by macroeconomic activity rather than firm-level operating performance.

4.2.2 Regional Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
CSI500	0.136**								0.062
	(2.17)								(0.68)
Debt		4.660***							5.230***
		(3.02)							(3.39)
Metal			0.123						0.044
			(1.39)						(0.38)
Energy				0.047					-0.172*
				(0.87)					(-1.67)
Airline					0.077				0.071
					(1.54)				(1.10)
Farm						0.148***			0.123*
						(2.72)			(1.77)
Coal							0.079*		0.121
							(1.69)		(1.35)
SPX								0.0715	0.072
								(0.98)	(0.98)
N	3402	3402	3402	3402	3402	3402	3402	3402	3402
Adj. R^2	0.001	0.002	0.000	-0.000	0.000	0.002	0.001	-0.000	0.005

Note: This table reports stepwise OLS regression estimates for the national carbon exchange sample. T-statistics are reported in parentheses. $*$ $p < 0.1$, $**$ $p < 0.05$, $***$ $p < 0.01$.

Table 5: Stepwise Regression Results – Regional

The regression results for the regional (non-national) carbon exchange in Table 5 reveal that four explanatory variables exhibit some degree of statistical significance in the individual regressions:

the CSI500 Index, the ChinaBond Composite Index, the Agriculture Index, and the Coal Futures Index. Among these, the bond and agriculture indices show the highest significance, with confidence levels exceeding 99%.

In the full model, the ChinaBond Index and the Agriculture Index remain significant, at the 1% and 10% levels, respectively—consistent with the full-sample results. Additionally, the Energy Futures Index shows a significant negative effect at the 10% level. This may reflect the mechanism through which rising energy prices increase firms' production costs, thereby reducing demand for emissions-intensive activities and suppressing carbon allowance returns.

Overall, the drivers of regional carbon allowance pricing align with those observed in the full sample, including sensitivity to bond market-implied transaction costs and the agriculture sector's emissions profile. However, the negative impact of energy prices is more pronounced in the regional sample, likely due to the structural differences in sectoral coverage and trading frequency between national and regional exchanges (see Table 1 and Figure 4). Accordingly, the following section focuses on the national exchange to further investigate its distinct pricing determinants.

4.2.3 National Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
CSI500	0.060								0.042
	(0.73)								(0.37)
Debt		-1.467							-1.053
		(-0.78)							(-0.55)
Metal			0.229**						0.269**
			(2.35)						(2.17)
Energy				0.000					0.111
				(0.00)					(0.85)
Airline					0.056				-0.015
					(1.07)				(-0.22)
Farm						0.057			0.041
						(0.84)			(0.49)
Coal							-0.040		-0.207*
							(-0.71)		(-1.71)
SPX								-0.003	0.008
								(-0.04)	(0.10)
N	552	552	552	552	552	552	552	552	552
Adj. R^2	-0.001	-0.001	0.008	-0.002	0.000	-0.001	-0.001	-0.002	0.004

Note: This table reports stepwise OLS regression results for the regional carbon exchange sample. T-statistics are reported in parentheses. $*$ $p < 0.1$, $**$ $p < 0.05$, $***$ $p < 0.01$.

Table 6: Stepwise Regression Results – National

The regression results in Table 6 reveal that the determinants of carbon allowance pricing in the national exchange differ substantially from those in the full sample and the regional exchange sample. In the full model, macroeconomic indicators such as the ChinaBond Composite Index, CSI500 Index, and S&P 500 Index show no statistically significant influence.

The most significant factor appears to be the Metals Futures Index, which shows a strong positive effect on carbon returns at the 1% confidence level. In addition, the Coal Futures Index exhibits a statistically significant negative effect at the 10% level in the full model. This may reflect a mechanism wherein rising metal prices enhance the profitability of metallurgical firms, thereby increasing their demand for emissions allowances and pushing up allowance prices. Conversely, higher coal prices raise production costs for carbon-intensive firms, which may reduce their emissions output and subsequent demand for carbon permits, thereby lowering prices.

Unlike the full and regional samples, the national exchange sample shows almost no sensitivity to broad macroeconomic conditions. Instead, it appears more responsive to industry-specific factors and energy market conditions. This divergence could be attributed to differences in sectoral coverage (see Table 1) and regional characteristics of the exchange itself, as the national carbon trading platform is headquartered in Shanghai, China.

4.3 Robustness Check

To address potential nonlinearity and scale effects that may arise from using arithmetic returns, we perform a robustness check by modifying the original model (1). Specifically, we replace the percentage excess return of carbon allowances with log excess return, defined as $r_t = \ln(P_t/P_{t-1})$. The adjusted regression model is specified as:

$$\begin{aligned} Carbon_LN_t = {} & \beta_1 \cdot CSI500_t + \beta_2 \cdot SPX_t + \beta_3 \cdot Debt_t \\ & + \beta_4 \cdot Metal_t + \beta_5 \cdot Airline_t + \beta_6 \cdot Farm_t \\ & + \beta_7 \cdot Energy_t + \beta_8 \cdot Coal_t + \varepsilon_t \end{aligned} \tag{2}$$

where $Carbon_LN_t$ represents the daily log excess return of China's carbon allowances.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
CSI500	0.130**								0.062
	(2.34)								(0.77)
Debt		3.180**							3.788***
		(2.35)							(2.76)
Metal			0.140*						0.083
			(1.82)						(0.84)
Energy				0.039					-0.144
				(0.84)					(-1.59)
Airline					0.071*				0.057
					(1.66)				(1.02)
Farm						0.138***			0.113*
						(2.87)			(1.85)
Coal							0.062		0.083
							(1.52)		(1.05)
SPX								0.061	0.061
								(0.97)	(0.97)
N	3954	3954	3954	3954	3954	3954	3954	3954	3954
Adj. R^2	0.001	0.001	0.001	-0.000	0.000	0.002	0.000	-0.000	0.003

Note: This table reports robust check results using stepwise OLS regressions. T-statistics are reported in parentheses. $*$ $p < 0.1$, $**$ $p < 0.05$, $***$ $p < 0.01$.

Table 7: Robustness Check – Stepwise Regression Results

The robustness regression is performed on the same 3,954 daily observations as the baseline model. As shown in Table 7, switching from percentage to log returns leads to a slight reduction in the model's explanatory power, with marginally lower t-statistics across variables compared to Table 4.

Nevertheless, the key findings remain consistent: the ChinaBond Composite Index and the Agriculture Index retain their statistical significance at the 1% and 10% levels, respectively. This consistency reinforces the reliability of the original specification and suggests that the relationship between these macro/sectoral factors and carbon allowance prices is robust to return transformation.

5 Conclusion and Recommendations

5.1 Key Findings

Internationally, ETSs show greater price volatility and GHG coverage compared to carbon taxes. Carbon prices follow a pro-cyclical pattern with significant regional variation.

In China, nine ETS platforms cover sectors like energy, metallurgy, aviation, and agriculture. Since 2021, the national ETS has become dominant, with unique pricing behavior compared to regional exchanges.

Regression analysis shows:

- Aggregate and local ETS prices correlate positively with bond and agriculture sector indices.
- National ETS prices are not significantly influenced by macro indicators, but respond to sector-specific (metals, coal) signals.

5.2 Policy Suggestions

- **Introduce carbon taxes and offset credit markets:** Carbon taxes broaden price coverage, improve liquidity, and support clean energy. A regulated offset market can monetize afforestation and renewables, boosting dual-purpose income and climate goals.
- **Enhance inter-provincial coordination:** Unifying trading rules and data collection can

improve efficiency and research quality.

- **Promote international alignment:** China can lead system harmonization efforts to reduce fragmentation and avoid carbon trade barriers.

5.3 Future Research

Future studies could explore socio-economic variables (e.g., education, income inequality) as additional determinants of cross-country carbon price variation. Analyzing these could refine understanding of behavioral and institutional factors shaping carbon pricing.

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